NYC Restaurant Health Grading Prediction

Section 003
Group 39
Jiabei Zhao, Zixun Chen, Zixuan Liu, Leo Lin, Raphael Monges

1. Introduction

The New York City Department of Health assigns letter grades to food establishments based on sanitary inspection scores, providing transparency for consumers and encouraging compliance with public health standards. This project aims to predict restaurant inspection outcomes using machine learning models by leveraging a comprehensive public dataset containing past inspection results, establishment characteristics, and contextual information. Our goal is to identify the key factors associated with poor sanitary performance and to develop models capable of anticipating future inspection grades. Such predictive insights can help regulatory agencies prioritize inspections, allocate resources more effectively, and support proactive interventions by identifying high-risk establishments before violations occur. To achieve this, we employ classification models including XGBoost and Neural Network to predict whether an inspection will result in an A, B, or C grade. The resulting predictions offer practical value for improving inspection planning, guiding food safety initiatives, and ultimately enhancing public health outcomes.

2. Data Cleaning and Preprocessing

Our original dataset contained **292,053 inspection observations** with **27 attributes**, describing restaurant identifiers, inspection dates, geographic locations, violation details, cuisine types, inspection types, and grading outcomes. Since many attributes were irrelevant for modeling, we perform substantial preprocessing and feature engineering. Our final analytic dataset consists of **11 engineered features** categorized into three main components: **Location-based characteristics, Restaurant/cuisine characteristics and Inspection timing & historical performance.** These refined features were selected based on domain knowledge, predictive relevance, interpretability, and data availability.

Because many inspections do not explicitly report letter grades but instead list **numeric sanitary scores**, we standardized outcomes by converting score values into three categorical grades consistent with NYC regulations: **A (score ≤ 13), B (14–27), and C (≥ 28)**. This allowed us to create a consistent and interpretable target variable across inspections with or without explicit grades. Observations missing score information were removed to ensure outcome reliability.

The raw dataset reported more than **30 inspection types**, which were too granular and sparse for predictive modeling. To improve interpretability and reduce noise, we consolidated these categories into six broader groups that reflect meaningful regulatory distinctions: **Cycle, Calorie Posting, Smoke Free, Pre-Permit, Inter-Agency, Trans Fat, and Other**. These aggregated categories maintain important differences in inspection intent while avoiding unnecessary fragmentation.

The dataset contained over **90 cuisine types**, many of which were extremely rare and introduced sparsity. To address this, cuisine categories with **fewer than 100 restaurants** were grouped into a single **"Other" category**, while the remaining cuisine labels were retained. This standardization preserves meaningful variation across food types while ensuring that rare or idiosyncratic categories do not distort the model.

Inspection performance often reflects **long-term patterns** rather than isolated events, so we engineered features that capture timing and history. We extracted the month and year of each inspection and computed the number of prior inspections for each restaurant. To summarize historical performance, we identified previous B and C outcomes as violations and calculated both the **cumulative number of violations** and **the violation rate** (PRIOR_VIOLATIONS / PRIOR_INSPECTIONS). These features help distinguish restaurants with persistent sanitation issues from those with occasional infractions.

After preprocessing, the final dataset includes the grade category (A/B/C), geographic attributes (borough, ZIP code, latitude, longitude), standardized cuisine and inspection-type categories, temporal indicators (month and year), and historical metrics such as prior inspections and violation rate. Together, these features provide a structured and informative foundation for the predictive models developed in subsequent sections.

3. Exploratory Data Analysis

After cleaning and filtering the data, we generated a series of visualizations that help contextualize the distribution of restaurant inspections, the grading structure, and historical performance patterns (graphs are shown in the Appendix).

Top 15 Cuisines in NYC Restaurant Inspections: American, Chinese, Coffee/Tea, Pizza, and Mexican cuisines dominate the inspection data, reflecting NYC's diverse food landscape. Different cuisine types might show distinct inspection patterns worth noting for modeling.

Number of Inspections by Borough: Manhattan, Brooklyn, and Queens lead in inspection counts due to their larger restaurant populations, while the Bronx and Staten Island have notably fewer inspections.

Level of Violation: Using NYC Health's scoring system, we split restaurants into A (0–13 points), B (14–27 points), and C (28+ points) grades. The distribution shows A and B are pretty evenly split at 37.6% and 34.9%, with C trailing a bit at 27.5%. This relatively balanced spread makes the grade categories a solid target for prediction.

Distribution of Score Categories by Borough: Staten Island shows the highest share of A grades (40.1%) and lowest C grades (26.3%), while Queens has the most C grades (38.9%). Brooklyn also has a notably high C proportion (36.1%). These borough-level differences suggest geography plays a role in hygiene outcomes and should be accounted for in modeling.

Prior Violation Rate vs Prior Inspections: This scatter plot connects a restaurant's inspection history with its violation track record. Restaurants that have been inspected more times in the past tend to show higher historical violation rates, pointing to consistent sanitation behaviors over time. This makes features like prior violation rate and inspection count really valuable for predicting future outcomes

Heatmap of Restaurant Locations in NYC: The density map confirms what we expect—restaurants cluster heavily in Manhattan, especially Midtown and Downtown, with significant concentrations in parts of Brooklyn and Queens too. This spatial layout matches the

city's restaurant reality and helps explain some of the geographic variation we see in inspection results.

4. Models

4.1 SVM:

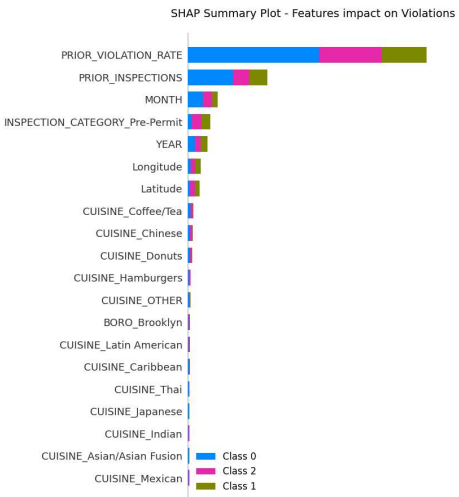
Since the main goal of the project is to determine the severity of the violations grouped into three distinct levels: A, B, and C, the problem is inherently a multiclass classification task. We therefore considered many different candidates that are suitable for multiclass classification like SVM, kNN, and Multinomial Logistical Regression. However, the feature set consisted of categorical variables like ZIP CODE and CUISINE, which after One-Hot Encoding, would lead to a large, high-dimensional sparse feature matrix. Distance-based methods like kNN are known to perform poorly under such sparsity, leading to degraded predictive performance. These considerations motivated our decision to focus on SVM-based approaches. To mitigate overfitting and improve generalizability, we employ 3-fold cross-validation as part of the model selection process. In the context of SVMs, a critical component of this process is kernel choice, which strongly influences the decision boundary and overall performance. By default, kernelized SVM in sklearn uses the radial basis function, which scales exponentially with number of samples and features, making the training process prohibitively slow with a large dataset (over 250,000 rows and over 200 columns after one-hot encoding.) For this reason, we chose LinearSVMs. Modern LinearSVM implementations rely on stochastic gradient descent based optimizations, and scale linearly with the number of samples and features O(Tnd), making it more suitable for our large, high dimension dataset.

LinearSVM achieved an average accuracy of 0.554 across the 3 folds. The model performed well when predicting A & C achieving recall of over 67%. However, the model performs poorly when predicting the “B” class, with recall of only around “33%.” This is likely because due to the borderline nature, the model does not predict “B” much. The only hyperparameter to be tuned in LinearSVM is C, the regularization parameter. We iterated over a selected amount of regularization (0.01, 0.1, 0.3, 0.5, 1.0) to find the optimal C of 0.5. We also found that it does not significantly impact the result, resulting in an average accuracy of 0.555.

4.2 XG Boost

We decided to implement XGBoost, it has the ability to model nonlinear relationships, and capture complex interactions among features. Our dataset includes categorical variables (borough, cuisine, ZIP code, inspection category) and numerical features such as temporal indicators.

Before training the model, we applied one-hot encoding to categorical features and standardized numerical variables to ensure scaling. To evaluate model performance, we used a 3-fold cross-validation approach, training an independent XGBoost classifier on two-thirds of the data and testing it on the remaining third in each fold. The goal is to reduce overfitting and get a better measure of generalization.

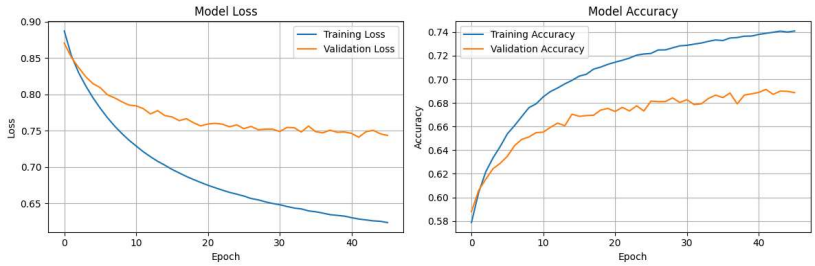


XGBoost achieved fold accuracies of 0.6653, 0.6614, and 0.6676, with a mean accuracy of 0.6648, indicating stable performance. The model performed well on Grade A and Grade C, with strong recall for both, while Grade B remained harder to classify due to its borderline nature.

To interpret the model, we used SHAP, which identified "prior_violation_rate" and "prior_inspections" as the most influential predictors. Temporal features and inspection type also contributed, whereas cuisine and location played smaller roles.

4.3 Simple Neural Network

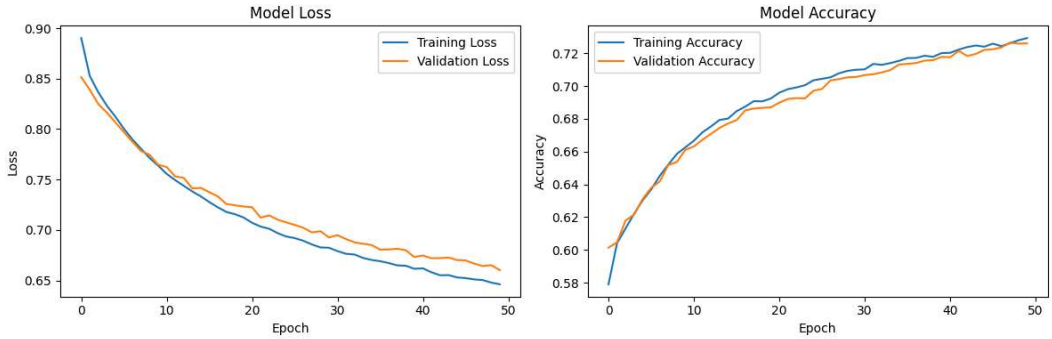
We also trained a neural network on the same engineered feature set. After one-hot encoding the categorical variables and standardizing numerical features, we split the data into training (80%) and test (20%) sets. The network consists of a single hidden layer with 128 ReLU units and a 3-node softmax output layer for the A/B/C grade classes. We trained the model with the Adam optimizer and categorical cross-entropy loss, using a batch size of 64 and early stopping based on validation loss to prevent overfitting.



The learning curves show that training loss decreases while validation loss plateaus after roughly 20-25 epochs. The gap difference between curves can suggest some overfitting but looks like a good generalization. Training accuracy approaches about 0.74, while validation accuracy stabilizes around 0.68–0.69, comparable to the performance of the XGBoost model. Overall, the neural network captures nonlinear patterns in the data but does not substantially outperform the gradient-boosted trees, while offering less interpretability.

4.4 Deeper Neural Network

We trained a deeper neural network on the same processed feature set. The model takes the full engineered feature vector as input and passes it through three hidden layers of 250 ReLU units each, with Dropout(0.3) after every hidden layer to regularize the network and mitigate overfitting. As before, the output layer is a 3-node softmax corresponding to grades A, B, and C. We trained the model with the Adam optimizer and categorical cross-entropy loss, using a batch size of 64, a 20% validation split, and early stopping based on validation loss. The learning



curves show training and validation loss decreasing smoothly and in parallel, while accuracy rises for both sets, suggesting good generalization rather than severe overfitting. On the held-out test set, the deeper network achieved a test accuracy of 0.7248 with a test loss of 0.6630, outperforming both the simpler neural network and the XGBoost model. It can indicate that our regularization was helpful and that a deeper architecture can better capture the complex, nonlinear relationships underlying restaurant inspection outcomes.

5. Conclusion:

The analysis is based on restaurant statistics published by NYC Department of Health, aiming at using machine learning models to predict the restaurants' health level (A, B or C). The predictions can be helpful to distinguish health-risky restaurants and the analysis results can reveal some key factors impacting restaurant health level. According to the exploratory data analysis part, distribution of health inspection results of nyc restaurants have significant differences with different cuisine types, spatial layouts and boroughs. Also, there is a high correlation between historical violations rate and restaurant prior inspections, indicating an existence of consistent sanitation behaviors of restaurants.
For the modeling part, there are a total of four parts: SVM, SG Boost, Simple Neural Network and Deeper Neural Network. LinearSVM has an advantage on high dimensional calculation but its performance on predicting "B" is not as good as expected. SG Boost has a shocking overall prediction ability and through SHAP it identified "prior_violation_rate" and "prior_inspections" these influential predictors. Neural Network models enhanced the analysis by optimizing in non-linear relationships. By theory Neural Network models should have the strongest ability in test accuracy, and the analysis results match the theory (highest accuracy: 72.48% among all the models).

6. Possible Improvements

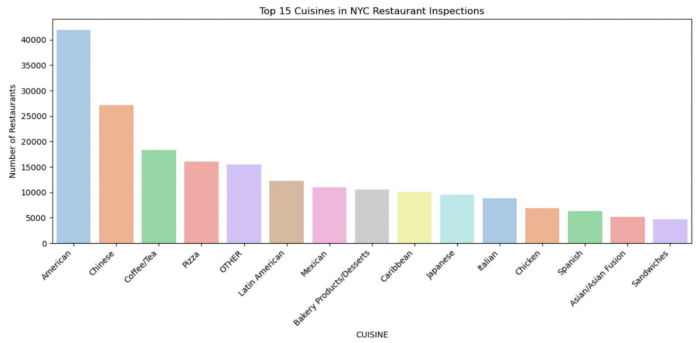
"prior_violation_rate" and "prior_inspections" played an important role in the entire research and analysis, however, long-period historical data may not be as important as expected. In the research, treating old data and recent data in the same way may cause some potential problems. Adding weights to historical data is one of the possible improvements for the research. If the data is too old, it should not be a good reference thus the weight will not be as high as recent data. Besides, the analysis part contains factors like borough and spatial layouts, it may cause some bias in location. Future research can pay attention to examining the fairness related to possible-biased variables.

References:

https://a816-health.nyc.gov/ABCEatsRestaurants/#!/faq

Appendix：

3.1



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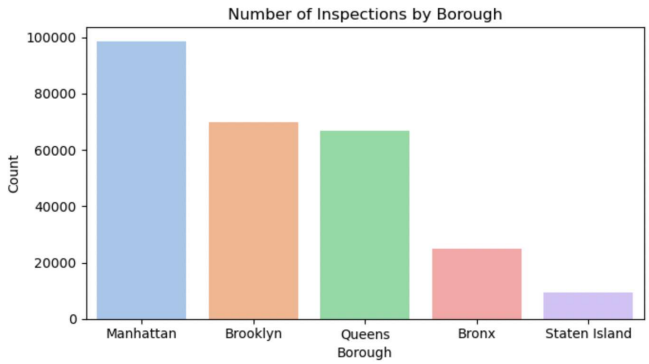

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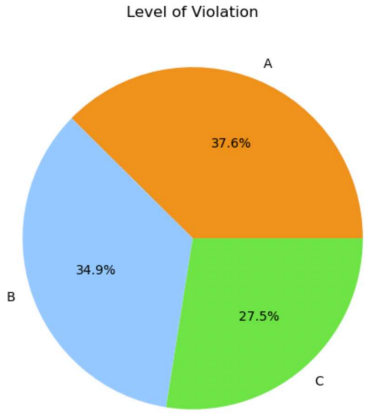

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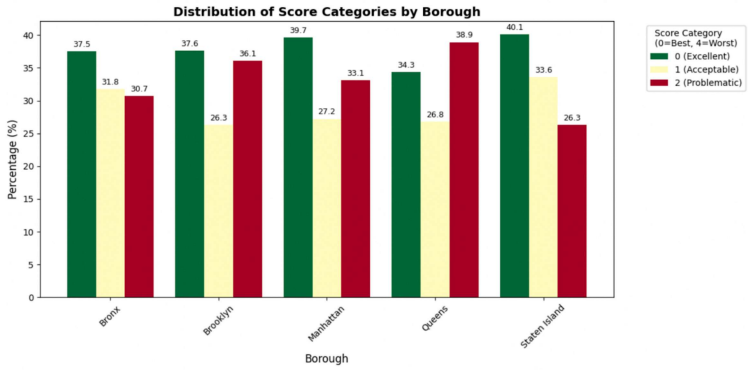

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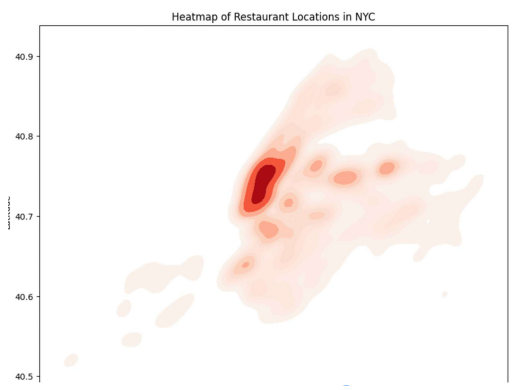

Confusion Matrix for LinearSVM

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[[21491   724 10951]
[13028   881 10645]
[12999   667 17314]]
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